

Time and Gravity Research

GR and QFT connection

Time, Entropy, Global Time, Local Time, Proper time and Gravitational Time Dilation Mechanisms

-The Quantum Thermodynamics Bridging Thermodynamic Time, Relativistic Time and Quantum Field Theory Time Parameters

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Abstract: This study presents a novel discrete model that unifies time and entropy through a spacetime mapping framework. The model rigorously satisfies the thermodynamic arrow of time while maintaining consistency with conventional entropy principles. By establishing fundamental connections between global time (cosmic transformations), local time (regional spacetime dynamics), and thermodynamic evolution, it provides a computational foundation for physical simulations, and provides a physical mechanism explaining gravitational time dilation.

Key theoretical implications include deriving time's directionality from quantum spacetime network dynamics, explaining light-speed invariance, and proposing dark energy/matter as emergent phenomena of spacetime's ground state. A distinctive falsifiable prediction suggests a measurable difference between positron and electron magnetic moments due to chiral quantum coupling.

The model offers three foundational constraints for computational physics: thermodynamic conservation laws, entropy-driven evolution, and least-action path selection. This framework bridges discrete spacetime dynamics with macroscopic phenomena while proposing testable quantum signatures.

Key words: multiplicative entropy; space-time-entropy mapping; thermodynamic time arrow; entropy coordinate; time dilation; global time; proper time; local time

Introduction:

Time has always stood as one of the deepest puzzles in both physics and human thought. From ancient philosophers to modern scientists, its relentless forward march has inspired countless theories, yet its fundamental nature remains elusive. Einstein's relativity transformed time from an absolute backdrop into a dynamic fabric, while thermodynamics gave us the arrow of time through entropy. But critical gaps persist between these understandings—why does time flow at different rates in different contexts? What truly governs its irreversible direction?

This work offers a new perspective by revealing time as an emergent property of spacetime's quantum structure. We propose that what we experience as time is fundamentally the manifestation of discrete transformations in the fabric of reality itself. Rather than treating time as an independent dimension, our model shows how it arises from deeper principles: entropy acting as the bridge between quantum-scale events and cosmic evolution.

Within this framework, phenomena like time dilation and the arrow of time are no longer separate mysteries, but different expressions of the same underlying mechanics. The speed of light becomes a natural consequence of spacetime's quantum network properties, while dark energy and matter emerge from its ground-state behavior. Crucially, this perspective makes concrete predictions, including subtle but measurable differences in particle behavior that could confirm or challenge the theory.

By distinguishing between global cosmic time and local relativistic time through global or local space transformations, this approach provides not just theoretical insight but a practical foundation for computational physics. It suggests that time, in all its manifestations—from the ticking of clocks to the expansion of the universe—may ultimately be understood as the dynamic expression of spacetime's most fundamental architecture.

I Basic set:

1.1 The universe is in a state of expansion.

1.2 The universe adheres to the Law of Energy Conservation and the Law of Entropy Increase.

1.3 The universe is fundamentally composed of Space Elementary Quanta (SEQ) - discrete Planck-scale units that form a topologically homeomorphic 3D structure. While individual SEQ energy states may vary, their adjacency relations remain strictly preserved. Crucially, this polycrystalline structure inherently contains primordial dislocations that ensure physical isotropy, yet maintains rigorous 3D topological homeomorphism as these defects have remained cosmologically frozen and adjacency-preserving since the universe's inception. The observed gravitational lensing of light around black holes provides compelling evidence for the coupling

between gravitational and electromagnetic field quanta, strongly suggesting they represent different excitation states of the same underlying quantum field - a key postulate of the SEQ hypothesis. Within this framework, both the spacing and tension between adjacent SEQs are dynamically modulated by gravitational or equivalent gravitational field interactions.

1.4 All field quanta and elementary particles represent distinct energy excitation states of SEQ, expressible as 3D dynamic structural matrices of SEQ.

1.5 SEQ possess a ground state energy, such as ground-state spin or vibrational modes. If the ground-state spin chirality is fixed, this could explain parity violation. Additionally, the ground-state energy of SEQ may account for the cosmological constant in General Relativity.

This framework aligns closely with Loop Quantum Gravity theory [1, 2].

1.6 Adjacent SEQs maintain a dynamic equilibrium spacing. Deviations from this equilibrium alter the ground resonant frequency of SEQs while generating corresponding compressive or tensile forces - the fundamental mechanism underlying gravitational phenomena. This response exhibits intrinsic nonlinearity and asymmetry, potentially manifesting as complex regional gradient variations. The elastodynamic description of spacetime in this framework demonstrates striking conceptual parallels with Einstein's elastic continuum analogy [3], while materializing Zee's "spring network" concept [4] through the discrete SEQ architecture. As fundamental Planck-scale constituents, SEQs preserve their structural integrity regardless of deformations or energy state transitions.

Within this framework:

- The spin degrees of freedom of SEQs remain dynamically decoupled from their elastic bonds
- System perturbations induce modifications to SEQ resonant frequencies alongside force generation
- The nonlinear, asymmetric response enables emergent phenomena like regional gradient variations
- SEQs constitute stable, indivisible structures with spin emerging from sub-Planckian collective excitations
- Sub-Planckian elasticity imposes fundamental limits on inter-SEQ spacing and tension modulation
- This architecture naturally isolates spin dynamics from elastic disturbances
- Extreme deformations (e.g., near black holes) cannot disrupt the network's topological integrity due to these fundamental constraints.

1.7 While matter appears to traverse through space, this motion fundamentally represents propagating excitations within the spacetime medium itself -a key postulate of this framework that matter is an integral part of spacetime. This

interpretation is consistent with General Relativity, as evidenced by black holes preserving local spacetime topology despite their extreme mass concentrations (manifested through smooth light-bending effects). Consequently, the speed of light (c) constitutes the maximum propagation rate for these spacetime excitations, establishing an absolute velocity limit that prevents superluminal stacking. This framework naturally incorporates Einstein's postulate of an invariant maximum speed while providing a discrete substrate for relativistic phenomena.

2. Definition of Time

2.1 In the SEQ framework, electromagnetic waves propagate through the SEQ medium itself. Massive matter and its motion appear as wave excitations within this medium. A core principle of this model states that no actual movement through space occurs - the speed of light c serves as the ultimate propagation limit, fundamentally preventing any velocity addition beyond c .

2.2 Universal composition: Energy conservation and quantization dictate a finite count N of SEQs, each with M discrete energy states (integer multiples of Planck's constant h), permitting up to M^N potential transformations. These energy states form an algebraic system encompassing translational, spin and rotational degrees of freedom. The joint constraints of energy conservation and entropy maximization dramatically restrict the accessible state space, reducing physically realizable transformations to a tiny fraction of the theoretical M^N possibilities.

2.3 Temporal definition:

2.3.1 Let J represent possible universal state transitions ($J \ll M^N$)

2.3.2 Consecutive transformations are separated by Planck time (t_p) intervals

2.3.3 The temporal arrow follows entropy increase

2.3.4 Transformations map non-bijectively to entropy values (k distinct values partition J into K classes)

2.3.5 Each space transformation carries a unique entropy value calculated via multiplicative energy distribution

2.3.6 Finite transformations ensure discrete, finite time set.

This discrete evolutionary approach shows conceptual parallels with causal set theory's treatment of spacetime dynamics [5], while differing fundamentally in its entropic time construction.

3 Entropy Definition and Analytical Formulation

The entropy in this framework is calculated using the energy state norms of Space Elementary Quanta (SEQ). This formulation enables the determination of entropy values for either localized closed systems or the universe as a whole. The fundamental definition expresses entropy as the multiplicative accumulation of energy state norms across the system under consideration.

Key features:

- Based on SEQ energy state norms
- Applicable to both local and global systems
- Defined through multiplicative accumulation
- Maintains consistency with thermodynamic principles

This multiplicative approach to entropy quantification provides a natural framework for discrete systems while preserving essential thermodynamic properties. The formulation demonstrates particular utility in modeling irreversible processes within the SEQ architecture.

$$\text{Entropy value of closed system } S = \prod_{i=1}^n m_i, i \in N \quad \textcircled{1}$$

$$\text{Energy of closed system} = \text{constant} = \sum_{i=1}^n m_i, i \in N \quad \textcircled{2}$$

$S_{\max} \leq m_i^n$, When all m_i are equal or differ only by Planck's constant h
(Where m_i refers to the energy carried by the i th SEQ during a single transformation of the closed system, where each energy state is an integer multiple of Planck's constant h ,)

3.1 Energy transfer rules and triggering conditions:

Energy exchange occurs between adjacent SEQ (i, j) if and only if the following

thermodynamic gradient exists: $m_i > m_j + h$, Energy transfer occurs only in discrete

quanta of Planck's constant h , $m_i \rightarrow m_i - h$; $m_j \rightarrow m_j + h$ (Planck's constant: h)

3.2 Logarithmic Relation:

After logarithmic transformation, $\ln S$ aligns with the conventional Boltzmann entropy form, while the multiplicative formulation naturally suits discrete systems.

3.3 Proof of Spontaneous Entropy Increase

Theorem:

For every possible energy transfer process, the total entropy change satisfies $\Delta S \geq 0$.

Proof Outline: Let the pre-transfer states be $m_i = a$, $m_j = b$ ($a > b + h$);

The post-transfer entropy ratio is:

$$S_{t2}/S_{t1} = (a-h)(b+h)/ab = 1 + h(a-b-h)/ab > 1 \text{ (Planck's constant } h)$$

3.4 Numerical Example: System States and Entropy Evolution

Path 1: Maximum Entropy Path

Step	Energy Distribution	Entropy (Πm_i)	Entropy Change
0	[3, 1, 5, 3]	45	-
1	[3, 2, 4, 3]	72	+27
2	[3, 3, 3, 3]	81	+9

Result: Reaches maximum entropy (81) with uniform distribution in 2 steps.

Path 2: Non-Maximum Entropy Path

Step	Energy Distribution	Entropy (Πm_i)	Entropy Change
0	[3, 1, 5, 3]	45	-
1	[3, 1, 4, 4]	48	+3
2	[3, 2, 3, 4]	72	+24

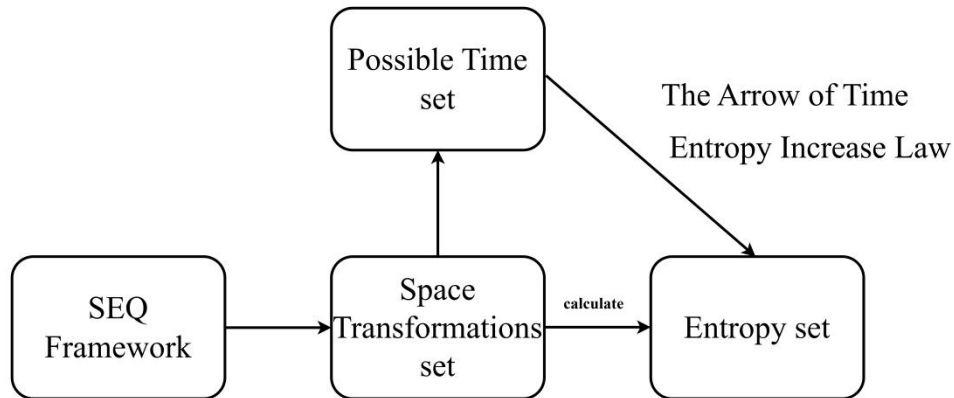
Result: Stops at entropy=72 (non-uniform), as no further transfers satisfy $m_i \geq m_j + h$. The above examples clearly demonstrate the differences in entropy increase across different evolutionary paths.

3.5 Entropy increase manifests through macroscopic interactions level, atomic/molecular level, QED processes, and microscopic particle restructuring (matrix disintegration/reorganization, radiation).

3.6 The entropy formulation $= \prod_1^n m_i; i \in N$ proposed herein operates not only at the fundamental SEQ level, but also admits coarse-graining for arbitrary massive many-body systems. This universality originates from the energy-mass equivalence which allows the cumulative product to naturally encode multi-body interactions. Crucially, this formulation provides first-principles computational constraints for discrete simulations across all scales.

At this point, a clear multiple mapping can be established:

Space-Time-Entropy Mapping Diagram



Draw tool: draw.io [Computer software]. <https://github.com/jgraph/drawio>

Diagram 1: Space-Time-Entropy Mapping Diagram

Drawing tool: draw.io [10]

4 Analysis of Action

The dimension of the action quantity is consistent with that of Planck constant, so the action quantity can be the number of units of quantum energy (Planck constant).

The total number of unit quantum conduction energy of a physical process involves two parameters, one is the number n of SEQ involved in the physical process wave conduction, and the other is the conduction times k_i of the i th space elementary quantum involved in the conduction, in which the maximum number of conduction count k_i of all single SEQ is less than or equal to times of local spatial transformations in this process. h is Planck constant. Then the action amount can also be written as

$$\sum_{i=1}^n h k_i, i \in N \quad (3)$$

The law of the principle of least action reveals that the path selected by physics process is the path that the total amount of energy conduction involved SEQ is the least. It involves two parameters, as mentioned above. If the number of SEQ involved remains the same and the conduction times of each SEQ are the same, (1)Fermat's principle of the shortest time of optical path and (2)the principle of the steepest descent line can be directly derived, because in these two examples, the number of SEQ involved in the physical process and the rolling spherical rigid body in the steepest descent line remain the same, and the conduction times of each SEQ are also approximately the same, then time (the number of local space transformations) is the only variable for the calculation of action, so using time to divide the wave forms of different paths is equivalent to using action to divide the wave forms of different paths. Time here refers to the transformation times of local space. From this point of view, we can probably understand why the analysis of action amount proposed in different

periods is different, but it can explain some phenomena.

5 Local time , the proper time, time dilation and relative time in Relativity

5.1 local time. As previously established, global time is defined by the transformations of the universe. This work introduces local time as an operational concept that: (1) provides correspondence with special relativistic time notions; and (2) enables precise specification of time scales for localized physical processes. Crucially, any measurable time parameter in physical calculations necessarily corresponds to transformations within a specific local space. This operational concept is designated as **local time**.

The local space scope must be unambiguously specified: either as the SEQ network along the physical process path or the connected region within the observer's light cone. This distinction mirrors the complementarity between Feynman's path-integral formulation and relativistic theory, avoiding conceptual confusion in prior works.

Local time: it can specify the time set corresponding to the transformations within a specific spatial range. In fact, from the perspective of this framework, the existing equations with t parameter in textbooks are actually the local time by default.

One example: the clock slowness effect in Relativity manifests as different transformation times in different local spaces.

Every local space transformation constitutes a part of the universal evolution. Global time progression (ΔT_{global}) does not necessitate synchronous local transformations, the state matrix of a given local space may remain invariant despite cosmic-scale changes ($\Delta \tau_{\text{local}} \approx 0$) when this local space undergoes no state transition.

5.2 The proper time in Relativity is related to local transformations count of physics process entities.

5.3 Time dilation: An object's motion (velocity/acceleration) deforms the SEQ network via: (1) stress-energy dependent spacing variation, (2) strain-modulated transformation frequencies, and (3) GR-consistent effects (velocity-dependence mirrors SR, acceleration induces curvature. The phenomenon of gravitational and equivalent gravitational time dilation stems from the modulation of local space transformation frequencies induced by curvature and tensor variations in spacetime.

5.4 Understanding on Lorentz-covariant rules in Special Relativity theory: The time perception of physical processes across distinct reference frames fundamentally corresponds to the observation of transformation counts. The observation time discrepancy between frames derives from the accumulated difference in their frame SEQ transformation counts. An observer measures another frame's time evolution through the differential transformation count ΔN , while he can't perceive their

own transformation count N_0 . The observed ΔN is fundamentally governed by the dynamic light-path difference between the observer's frame and the moving reference frame of the measured object. Under the principle of non-additivity of light speed (c-invariance), this formulation naturally derives Lorentz-covariant rules through counting operations.

- Key Distinction from GR Effects

While the mathematical derivation process aligns with standard special relativity textbooks—replacing continuous spacetime metrics with discrete counting operations—the physical interpretation differs substantially in conceptualization:

- SR Effects as Perceptual Phenomena

Time dilation and length contraction emerge purely as observer-dependent measurement consequences

Originate from information transmission constraints via discrete light-signal counting

- Contrast with GR Mechanisms

Gravitational time dilation involves actual deformation of the SEQ network's transformation frequency

Equivalence principle effects require space compression as well.

6 Phenomenological consistency checks

6.1 Why can't the speed of light stack up?

As established above, the speed of light (c) constitutes the maximum excitation propagation rate in space, wherein all observed motion fundamentally represents state transitions within this immutable lattice rather than physical traversal through space.

6.2 Uncertainty Relation and Wave-particle Duality.

The Uncertainty Principle naturally arises from wave propagation through discrete SEQ: precise position inherently limits determination of conduction speed (wave dynamics), and vice versa.

Wave-particle Duality. The wave nature is fundamental, while the particle nature emerges from the discreteness of spacetime itself.

6.3 Double-slit experiment. [6][7]

Explanation: Based on this framework, electron's conducting in the space conforms some probability function, and really causes multi-path oscillating in space, and these oscillating can be accumulated. Electron excitation causes space oscillating both of the two slits at the sides of excitation source, The proposed framework predicts the emergence of interference fringes, as the electron excitation induces coherent oscillations across multiple SEQ conduction paths. Even in case of one-slit experiment, when the accumulated Wavelets cross the slit, slit's Unsmooth edges may produce different paths and generate interference that could be observed if the sensor is sensitive enough and the slit edge's form can meet the Coherent Condition.

Note: While this framework strictly enforces time ordering along entropy increase, it currently does not provide an interpretation for delayed-choice experiments. This remains an open question requiring further development in this framework.

6.4 Non-conservation of parity

With the assumption above, if SEQ ground state spin chirality is fixed, that could be one possible explanation of the non conservation of parity.

Is dark matter potentially explained by high-density SEQ ground-state clusters under gravity? And is the ground state energy carried by the SEQ the so-called dark energy?

6.5 Conjecture on Muon Decay experiment[8]

An object's motion (velocity/acceleration) deforms the SEQ network via: (1) stress-energy dependent spacing variation, (2) strain-modulated transformation frequencies, and (3) GR-consistent effects (velocity-dependence mirrors SR, acceleration induces curvature). Particle decay results from destabilization of their 3D structure matrices when interaction forces (EM/weak/strong) fail to maintain equilibrium.

7 A prediction of a difference in the magnetic moments of the positron and electron(falsifiable)

Given the isotropic nature of the electric field generated by electrons, this framework hypothesizes that electrons possess an Spatially Symmetric structure composed of SEQ. Under the SEQ framework, all charged microscopic particles including electrons and quarks possess dynamic three-dimensional structures that preserve spherical symmetry in space.

If a slight difference is measured between the magnetic moment of the positron and that of the electron, it would strengthen the credibility of the hypothesis that the SEQ has a ground-state spin. This difference arises because the positron's structural matrix rotates with opposite chirality to the electron's, resulting in distinct coupling configurations with the SEQ's fixed-chirality ground-state spin. Based on this, it can be inferred that the magnetic moments of the positron and electron should exhibit a slight discrepancy. A precision measurement of the positron magnetic moment, currently underway by the Fan-Myers-Sukra-Gabrielse collaboration, could test two key hypothesis of the SEQ framework: the fixed chirality of SEQ ground-state spin and electron's substructure.[9]

8 Results

8.1. The study successfully establishes a time-entropy mapping model within the discrete spacetime framework, where each temporal state corresponds to a unique entropy value. This mapping ensures that entropy either increases or remains constant over time, providing a clear thermodynamic arrow of time.

8.2. However, the mapping is not bijective.

8.3. The model has undergone consistency checks against most key phenomena, confirming its physical plausibility, while proposing a falsifiable criterion.

9. Discussion

While the present model does not yet incorporate electromagnetic interactions or deeper algebraic structures, it establishes a foundational framework for such future developments. We envision that further development using an algebra system and discrete differential geometry could refine this framework, ultimately achieving better alignment with both general relativity and the Standard Model.

Within this framework, gravitational interaction emerges as a translational operation that modulates the inter-SEQ spacing and consequently alters the stress-energy distribution; the emergent gravitational field of mass exhibits spherical symmetry due to $SU(3)$ color charge distribution[11]. Serving as the symmetry-preserving mechanism at quantum scale, While this manuscript focuses on space-time-entropy mapping.

This work builds upon Wolfram' s foundational insight[12] that simple computational rules can give rise to complex emergent behavior. Where prior automata focused on pattern generation, our model demonstrates how such discrete systems can also encode the causal structure of physical laws—particularly the irreversible direction of thermodynamics and the path-selection principles of classical dynamics. This alignment between computation and physics reinforces the broader paradigm that nature' s behavior may indeed be rooted in elementary rules, provided those rules are constrained by fundamental invariants like energy conservation and entropy increase.

10. Conclusions:

This work presents a theoretical framework whose mathematical formulation, while deliberately minimalist, yields tractable derivations and falsifiable predictions. The model demonstrates particular utility in elucidating the fundamental relationship between temporal evolution and entropy, with immediate applications to computational simulations through the embedding constraints of thermodynamic laws and least-action path selection offering an entropy coordinate for computer evolution simulations.

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